Sharp Coordination Thresholds in Six-Matching Covers

Exact direct-repair widths, realizable extremal hypergraphs, and plateau obstructions

Guillaume Sidoine
Independent researcher
ORCID 0009-0005-4525-1182

Expanded research draft, 8 October 2026 — not submitted or externally refereed

Abstract

Consider six perfect matchings of a cubic graph and let Q be the sum of squared deviations of edge multiplicities from two. We distinguish strict defect descent from exact one-step correction and characterize the latter on every cubic bipartite graph: the minimum number of matching coordinates requiring replacement is a multicover transversal number. When all multiplicities belong to $\{1, 2, 3\}$, the transversal hypergraph is 3-uniform on six labels. In this regime, the minimum positive defects at which exact correction can require at least two, three, and four coordinates are sharply 6, 12, and 40. The 12 and 40 thresholds are realized by explicit 18- and 40-vertex 3-edge-connected cubic bipartite graphs with independently verifiable certificates. The $Q = 40$ example can nevertheless strictly improve with only two coordinates, underscoring the distinction between coordination width and descent. We also recover a sharp one-coordinate local plateau at $Q = 6$ and prove low-defect coordinated-descent bounds, as well as lifting the former obstruction to unbounded order and girth. None of the nonbipartite covering conjectures is resolved; novelty claims remain subject to specialist literature review.

1 Set-up and elementary barriers

Let G be a finite simple cubic graph with n vertices, and let $\mathcal{M} = (M_1, \dots, M_6)$ be an ordered six-tuple of perfect matchings. Put $m(e) = |\{i : e \in M_i\}|$ and

$$Q_G(\mathcal{M}) = \sum_{e \in E(G)} (m(e) - 2)^2.$$

Since each perfect matching covers $n/2$ edges and G has $3n/2$ edges, $\sum_e (m(e) - 2) = 0$. At each vertex, the three incident multiplicities sum to six. $Q = 0$ means that every edge occurs exactly twice. A *one-coordinate local minimum* means that replacing any one M_i by any perfect matching cannot strictly reduce Q. A *k-coordinate correction* changes at most k entries of the ordered tuple, with no restriction on the distance between the old and new perfect matchings.

Lemma 1.1. *For every simple cubic graph, if $0 < Q \le 4$, some one-coordinate replacement strictly decreases Q. Hence every positive one-coordinate local minimum has $Q \ge 6$.*

Proof. Write $d(e) = m(e) - 2$. At each vertex the incident values of d sum to zero. Modulo two, $Q \equiv \sum_e d(e) = 0$, so Q is even. If a nonzero edge has $|d(e)| \ge 2$, its two endpoints each force at least two additional units of squared defect among their other incident edges. Simplicity ensures these are four different edges; thus $Q \ge 4 + 4 = 8$. Consequently for $0 < Q \le 4$ every nonzero defect is ± 1, and at each affected vertex one $+1$ and one -1 edge meet. The defect edges therefore form disjoint cycles with alternating signs. Each cycle has

length at least four; thus $Q = 4$ and its support is an alternating 4-cycle. The two $+1$ edges each occur in three matchings. Let their sets of matching indices be A and B, both of size three. A -1 edge joining endpoints of these two positive edges occurs in a matching whose index lies outside $A \cup B$; therefore $A \cap B \neq \varnothing$. Choose $i \in A \cap B$. On the 4-cycle, M_i contains both positive edges and neither negative edge. Flipping this alternating cycle in M_i produces a perfect matching and reduces Q to zero. □

2 An exact two-coordinate frontier

Proposition 2.1 (Frozen-factor completion)**.** *Let G be d-regular and bipartite, and let $t, d \geq 1$ be integers. Take dt perfect matchings and fix a subcollection F of them. The fixed matchings can be completed to a dt-tuple in which every edge occurs exactly t times if and only if no edge occurs more than t times in F.*

Proof. Necessity is immediate. For sufficiency put $r(e) = t - \sum_{M \in F} \mathbf{1}_M(e) \geq 0$. At each vertex, $\sum_{e \ni v} r(e) = dt - |F|$, so the multigraph formed by $r(e)$ parallel copies of edge e is regular and bipartite. Kőnig's edge-colouring theorem decomposes it into $dt - |F|$ perfect matchings, as required. □

Lemma 2.2 (Small transversals for six matching labels)**.** *Let $S_1, \ldots, S_r$ be subsets of a six-element set, each of size at least three. If $r \leq 5$, a two-element set meets every S_j. If $r \leq 19$, a three-element set meets every S_j.*

Proof. Every S_j excludes at most three two-element subsets, since its complement has size at most three. There are 15 pairs in total. If $r \leq 4$, at least one pair is not excluded. If $r = 5$ and all pairs were excluded, equality in the counting bound would force the complements of the five S_j to be five three-element sets partitioning the edge set of K_6 into triangles. This is impossible because each vertex of K_6 has odd degree five, whereas any edge-disjoint union of triangles has even degree at each vertex. Thus a pair exists also when $r = 5$. Likewise, each S_j excludes at most one three-element subset. Since there are 20 triples, if $r \leq 19$ at least one triple meets them all. □

Theorem 2.3 (Bounded low-defect coordinated descent)**.** *Let G be a cubic bipartite graph and $\mathcal{M}$ a six-matching state with $Q > 0$. If $Q \leq 10$, a simultaneous replacement of at most two matchings strictly decreases Q. If $Q \leq 38$, at most three matchings suffice. In either case such a replacement can be found by polynomial-time bipartite matching computations (with at most six matching labels).*

Proof. Put $d(e) = m(e) - 2$. Because $\sum_e d(e) = 0$,

$$Q = \sum_e d(e)^2 \geq \sum_e |d(e)| = 2 \sum_{e : d(e) > 0} d(e) \geq 2|\{e : d(e) > 0\}|.$$

Orient surplus edges $d(e) > 0$ from the left to the right bipartition and deficit edges $d(e) < 0$ from right to left. At every vertex incident with a defective edge, the equation $\sum_{e \ni v} d(e) = 0$ guarantees at least one incoming and one outgoing directed edge. Thus the defect digraph contains a directed cycle C with alternating surplus and deficit edges. There are at most $Q/2$ surplus edges in the entire graph. Each is present in at least three of the six matching coordinates. When $Q \leq 10$ (respectively $Q \leq 38$), Lemma 2.2 provides two (respectively three) matching indices whose union contains every surplus edge on C. Let k be the number selected and let $x(e)$ be their current aggregate multiplicity. Change $x(e)$ to $x(e) - 1$ on surplus edges of C and to $x(e) + 1$ on deficit edges of C, leaving all other edges unchanged. This is nonnegative by the hitting condition and preserves degree k at every vertex, since C alternates.

The modified integer multiplicity vector is therefore a k-regular bipartite multigraph and decomposes into k perfect matchings by Kőnig. Replacing only the selected matchings realizes this modification. For every surplus or deficit edge on C, the absolute value of the original defect drops by one; thus Q strictly decreases. Computationally, the hitting set can be found by testing the fifteen pairs or twenty triples of six labels, and the factor decomposition uses k bipartite matching computations. □

Theorem 2.4 (Sharp $Q = 6$ coordination threshold)**.** *For every simple cubic bipartite graph G, every six-matching state with $Q = 6$ admits a 2-coordinate correction reaching $Q = 0$. There exists a 3-edge-connected cubic bipartite graph and a $Q = 6$ state for which no 1-coordinate correction reduces Q.*

Proof. The argument in Lemma 1.1 shows that $Q = 6$ cannot contain an edge with $|d(e)| \geq 2$. Hence the support of d consists of one alternating 6-cycle. Its three positive edges each occur in exactly three of the six matchings. By Lemma 2.2, choose two matching indices that together meet all three positive edges. Freeze the other four matchings. Every edge outside the cycle has total multiplicity two, every negative edge has total multiplicity one, and on each positive edge at least one copy lies in the two unfrozen matchings. Thus no edge is used more than twice among the frozen four matchings. Proposition 2.1 with $(d, t) = (3, 2)$ completes the frozen four to an exact twice-cover by changing only the selected pair. The second assertion is proved by the explicitly certified example below. □

Proposition 2.5 (Nonbipartite $Q = 6$ pair criterion)**.** *For a six-matching state on any simple cubic graph with $Q = 6$, a direct two-coordinate correction to $Q = 0$ exists if and only if there are two matching indices that together meet all three surplus edges, and after freezing the other four matchings the residual multiplicity vector decomposes into even cycles (counting doubled edges as cycles of length two).*

Proof. For a chosen pair I, let $f(e)$ count its uses among the other four (frozen) matchings, and put $r(e) = 2 - f(e)$. The vector r is nonnegative exactly when I meets every surplus edge; this uses the alternating-six-cycle structure at $Q = 6$. Independently, $\sum_{e \ni v} r(e) = 2$ at every vertex. Thus the residual is a spanning degree-two multigraph. It is the union of two perfect matchings if and only if all its cycle components have even length. This equivalence does not require bipartiteness of the ambient cubic graph. □

Proposition 2.6 (Exact two-coordinate block optimization)**.** *For any fixed pair of indices in a sextuple of perfect matchings of a bipartite cubic graph, the best simultaneous replacement of those two matchings minimizing Q can be found in polynomial time by an integer minimum-cost flow.*

Proof. Freeze four matchings and denote their edge multiplicities by $f(e)$. Choose a new residual multiplicity $x(e) \in \{0, 1, 2\}$ such that $\sum_{e \ni v} x(e) = 2$ at every vertex, minimizing $\sum_e (f(e) + x(e) - 2)^2$. For each edge, the incremental costs of the first and second residual copies are $2(f(e) - 2) + 1$ and $2(f(e) - 2) + 3$, respectively. Encode these as two unit-capacity parallel marginal arcs in a bipartite transportation network requiring throughput two at each vertex. Its integer minimum-cost flow is optimal; the increasing marginal costs ensure the second copy is never chosen without the first in an optimal flow. The resulting degree-two bipartite multigraph splits into two perfect matchings by Kőnig's theorem, giving an optimal pair of replacement coordinates. □

3 The 18-vertex certificate

The bipartition is $L = \{0, \ldots, 8\}$ and $R = \{9, \ldots, 17\}$. Number the following edges in the order shown, starting from index 0:

$$
\begin{array}{lll}
(0,11),(0,14),(0,9) & (1,14),(1,13),(1,11) & (2,15),(2,17),(2,13)\\
(3,16),(3,11),(3,17) & (4,9),(4,15),(4,12) & (5,13),(5,12),(5,10)\\
(6,17),(6,10),(6,15) & (7,12),(7,16),(7,14) & (8,10),(8,9),(8,16)
\end{array}
$$

Represent matchings as 27-bit nonnegative integers using these edge indices. The initial ordered six-tuple is

$$(76093521, 26264228, 19173986, 70392073, 38347922, 21308684).$$

Its six defective edges are, with multiplicities,

e	$(4,9)$	$(4,12)$	$(5,12)$	$(5,10)$	$(8,10)$	$(8,9)$
$m(e)$	3	1	3	1	3	1

so $Q = 6$. The path obtained by replacing the third matching by 36053794, followed by replacing the sixth matching by 21283916, has potentials $6 \to 6 \to 0$.

For an independent universal certificate of local optimality, fix i and put

$$w_i(e) = 2(m(e) - 2) + 1 - 2\mathbf{1}_{M_i}(e).$$

For any other perfect matching N one checks exactly that

$$Q(M_1, \ldots, N, \ldots, M_6) - Q(\mathcal{M}) = \sum_{e \in N} w_i(e) - \sum_{e \in M_i} w_i(e). \qquad (*)$$

The supplementary JSON file supplies, for each i, integers $a_i(v)$ at all eighteen vertices such that

$$a_i(u) + a_i(v) \le w_i(uv) \quad (uv \in E), \qquad a_i(u) + a_i(v) = w_i(uv) \quad (uv \in M_i).$$

Summing the inequalities over any perfect matching N and using the equalities for M_i proves that $(*) \ge 0$. These six dual witnesses are each 18 integers; the `standalone_checker.py` file checks all inequalities using only Python's standard library. It also verifies that the graph stays connected upon deleting any zero, one or two edges, certifying 3-edge-connectivity. A second implementation exhaustively enumerates all 47 perfect matchings and finds minimum replacement change zero in every coordinate.

4 Exact direct repair and a sharp hierarchy

For a state $\mathcal{M}$ of a cubic bipartite graph, write $S_e = \{i \in [6] : e \in M_i\}$ and define $\kappa(\mathcal{M})$ to be its *exact direct-repair width*: the smallest number of whole matching coordinates that can be simultaneously replaced to attain $Q = 0$. This is emphatically different from the number needed for a strict improvement, which may be smaller.

Theorem 4.1 (Exact coordination criterion)**.** *For every sextuple of perfect matchings in a cubic bipartite graph,*

$$\kappa(\mathcal{M}) = \min\Big\{|I| : I \subseteq [6], \quad |I \cap S_e| \ge \max(0, m(e) - 2) \text{ for every } e\Big\}.$$

In particular $\kappa(\mathcal{M}) \le 4$, and an optimal direct repair can be computed in polynomial time.

Proof. Fix a proposed set I of coordinate indices and freeze the others. If $|I \cap S_e| < m(e) - 2$ for some edge, the frozen coordinates already use this edge more than twice, so completion is impossible. Conversely, under these inequalities every frozen edge multiplicity is at most two. Proposition 2.1 with $(d, t) = (3, 2)$ constructs the required replacement matchings by decomposing a regular bipartite residual multigraph. Test the 2^6 choices for I and factor the residual at the first feasible choice of minimum cardinality. Four coordinates always suffice because any two frozen matchings use an edge at most twice. □

Corollary 4.2 (Unit-defect hypergraph)**.** *Suppose* $m(e) \in \{1, 2, 3\}$ *for every edge and* $Q > 0$*. Let* $\mathcal{F} = \{S_e : m(e) = 3\}$ *be the multiset of surplus-membership triples on* $[6]$*. Then* $\kappa(\mathcal{M}) = \tau(\mathcal{F})$*, the minimum size of a transversal of this hypergraph, and* $Q = 2|\mathcal{F}|$*. In particular:*

$$Q \le 10 \implies \kappa \le 2, \qquad Q \le 38 \implies \kappa \le 3.$$

These conclusions guarantee a direct correction to $Q = 0$*, not merely one strictly improving step.*

Proof. In this regime, every edge requiring removal has multiplicity three, hence needs one selected coordinate from its triple S_e. At each vertex the sum of multiplicity deviations is zero; thus surplus and deficit edges occur in equal number. Each has squared deviation one, giving $Q = 2|\mathcal{F}|$. The direct bounds follow from Lemma 2.2. □

Theorem 4.3 (Sharp unit-defect direct-repair thresholds)**.** *Restrict to finite simple cubic bipartite graphs and positive unit-defect states. For* $k = 2, 3, 4$*, the smallest possible value of* Q *for which* $\kappa \ge k$ *is respectively*

$$\boxed{Q_2 = 6, \qquad Q_3 = 12, \qquad Q_4 = 40.}$$

The three thresholds admit witnesses in 3-edge-connected *cubic bipartite graphs, on respectively* 18*,* 18*, and* 40 *vertices.*

Proof. The first lower bound follows from Lemma 1.1: when $Q < 6$, the only possible nonzero unit-defect state has one alternating four-cycle, and one matching flip reaches $Q = 0$. The previously certified 18-vertex $Q = 6$ state in Section 3 has no improving single-coordinate move, so its exact direct width is at least two and, by Theorem 2.4, is exactly two.

For $k = 3$, Corollary 4.2 rules out $Q \le 10$. The new 18-vertex graph encoded in `q12_cycle_blocker_18.json` has precisely six surplus-membership triples

$$012, \quad 013, \quad 234, \quad 045, \quad 145, \quad 235,$$

where, for instance, $012 = \{0, 1, 2\}$ and indices are zero-based. Direct inspection of the fifteen pairs of labels shows that no pair intersects all six triples. Therefore $\kappa \ge 3$. Any three labels except the complement of an occurring triple can be chosen suitably; in particular Lemma 2.2 for three labels yields $\kappa = 3$. All 27 edges and six perfect matchings are encoded in the supplement; an independent dependency-free checker verifies cubicity, bipartiteness, every deletion of at most two edges, and the masks. The same checker exhausts all 54 perfect matchings and finds that the optimum after *any* simultaneous pair replacement is $Q = 4$. An explicit path has potentials $12 \to 4 \to 0$, using a pair replacement followed by a single-coordinate replacement.

For $k = 4$, Corollary 4.2 rules out $Q \le 38$. The 40-vertex certificate `q40_cycle_blocker_40.json` contains precisely the 20 different triples $\binom{[6]}{3}$ as surplus memberships, one per surplus edge of one alternating 40-cycle. For any three indices $I \subseteq [6]$, the complementary triple $[6] \setminus I$ is disjoint from I and occurs among the surplus edges. Thus I cannot effect a direct exact correction: $\kappa \ge 4$. Theorem 4.1 gives $\kappa = 4$. A dependency-free verifier checks all six perfect matchings, the 40-cycle, the 3-edge-connectivity, and explicit corrections with potentials $40 \to 0$ by four indices and $40 \to 10$ by two. Hence this is not a counterexample to strict two-coordinate descent. □

A compact description of the $Q = 40$ **certificate.** Write $u_i = i$ and $v_i = 20 + i$, $0 \le i < 20$, for the two bipartition classes. Add the cycle edges $u_i v_i$ carrying a triple T_i and $v_i u_{i+1 \bmod 20}$ carrying a singleton $\{d_i\}$. Add the perfect matching of neutral chords $u_i v_{\pi(i)}$,

carrying the two-element membership set $[6] \setminus (T_i \cup \{d_{i-1 \bmod 20}\})$. The table below specifies $T_i, d_i, \pi(i)$ completely; the independent verifier checks that the edge memberships partition [6] at each vertex and that deleting any two edges preserves connectedness.

i	T_i	d_i	$\pi(i)$	i	T_i	d_i	$\pi(i)$
0	234	5	1	10	035	2	3
1	023	4	4	11	134	5	16
2	015	4	14	12	012	4	13
3	135	0	11	13	025	1	15
4	235	0	10	14	045	1	2
5	124	5	9	15	245	0	7
6	034	2	19	16	123	4	18
7	145	2	17	17	125	4	5
8	014	3	12	18	013	2	8
9	024	1	6	19	345	0	0

Assurance and limitations. The finite certificates in this section are checked without trusting the mixed-integer optimization programs used in their discovery. The general criterion is a straightforward application of Kőnig's theorem: only the sharp *realizable extremal hierarchy* is proposed as potentially novel. A specialist novelty audit is still required. The bound $\kappa = 4$ concerns exact *one-step* repair; the unresolved global conjecture concerns strict improvement with at most two matching coordinates.

5 Lifting a sharp plateau to an infinite family

Lemma 5.1 (Balanced vertex expansion)**.** *Let G be a simple cubic graph with a six-matching state $\mathcal{M}$ and let v have incident multiplicities $(2,2,2)$. Replace v with the 3-pole $K_{3,2}$, attaching the three former neighbours to its three degree-two vertices. The resulting cubic graph G' has a state $\mathcal{M}'$ with $Q_{G'}(\mathcal{M}') = Q_G(\mathcal{M})$. If $\mathcal{M}$ is a one-coordinate local minimum, so is $\mathcal{M}'$. This operation preserves simplicity, bipartiteness when present, and 3-edge-connectivity when present.*

Proof. Each of the three incident edges at v occurs in exactly two of the six matchings. Every matching chooses one incident edge. In $K_{3,2}$, the two ports not chosen externally must be matched to the two internal degree-three vertices. For the two matching labels choosing the same external port, select opposite internal assignments. Each of the six internal edges then occurs exactly twice; every other edge retains its old multiplicity.

Every perfect matching of G' restricts to a perfect matching of G by contracting the 3-pole: both internal hubs must be matched to ports, leaving exactly one external port matched. Replacing one lifted matching changes the original-edge contribution to Q by an amount at least zero whenever the original tuple is a local minimum. The initial internal contribution is zero, and every new internal contribution is nonnegative. This proves the claim.

The gadget adds four vertices and is bipartite with its three ports in the old vertex's part. To check 3-edge-connectivity, consider any cut of size at most two. If all three ports are on the same side but an internal hub is on the other side, at least three internal edges cross. If the ports are split, at least two internal edges cross; equality two forces both hubs with the majority ports. A cut of size at most two then has no external crossing edges, so contracting the gadget would yield a cut of size one in G (the minority port's external neighbour lies on the minority side), a contradiction. If the whole gadget lies on one side, the cut corresponds unchanged to a cut in G. Hence no such cut exists. □

Corollary 5.2 (Arbitrarily large, 3-edge-connected sharp traps)**.** *For every $k \geq 0$ there is a connected 3-edge-connected cubic bipartite graph on $18+4k$ vertices with six perfect matchings forming a positive one-coordinate local minimum of defect exactly $Q = 6$. The tuple admits a 2-coordinate correction to $Q = 0$; indeed the base $Q = 6 \to 6 \to 0$ sequence lifts unchanged along the family.*

Proof. The vertex 0 in the base graph has incident multiplicities $(2, 2, 2)$, and the two replacements in Section 3 do not change its matching choices. Apply Lemma 5.1; the internal hubs are again balanced and their incident matching choices stay fixed during both replacements, so the construction can be repeated indefinitely. The local minimum and defect persist, as does the indicated path. □

6 Lifting under finite covers

Proposition 6.1 (Covering stability of bipartite dual certificates)**.** *Let G be cubic and bipartite, and let $\mathcal{M}$ be a one-coordinate local minimum for Q. Every finite k-sheeted graph covering $p : H \to G$ inherits a one-coordinate local minimum $p^{-1}(\mathcal{M})$ with potential $kQ_G(\mathcal{M})$.*

Proof. The preimage of a perfect matching is a perfect matching, and edge multiplicities pull back along p, so the potential scales by k. For each coordinate i, the minimum-weight perfect matching problem with weights w_i has an integral bipartite assignment dual attaining the original matching's weight. Pull its vertex potentials back along p. The edge inequalities and the equalities on the lifted matching remain valid because every edge of H projects to an edge of G. Thus no lifted coordinate can strictly improve the potential. □

Corollary 6.2 (High-girth obstructions)**.** *There are connected bridgeless cubic bipartite graphs of arbitrarily large girth that possess positive one-coordinate local minima of the six-matching quadratic defect satisfying $Q = |V(G)|/3$.*

Proof. A finite connected graph has finite connected covers of arbitrarily large girth: the fundamental group is finitely generated free, hence residually finite; taking a finite-index normal subgroup excluding the finitely many nontrivial closed reduced walks of length at most a chosen bound produces such a cover. Apply the preceding proposition to the base certificate. Every edge of a finite cover of a bridgeless graph lies on a cycle (a lift of some positive multiple of a cycle downstairs), so the covers remain bridgeless. The base graph has 18 vertices and defect six; both quantities scale with the covering degree. □

An explicit connected 2-fold cover obtained by twisting edge indices 0 and 6 has 36 vertices, girth 6, edge connectivity 3, and a locally optimal tuple with $Q = 12$. The provided six lifted dual certificates check this example.

7 Scope, relation to prior work, and outlook

The frozen-factor completion criterion is an elementary application of Kőnig's theorem and is not presented as a new existence theorem for cubic bipartite graphs. Similarly, our coordinate changes may replace entire perfect matchings: they are not 4-cycle flips in the sense of standard perfect matching reconfiguration [2]. A recent treatment of toric ideals and 3-colour transformations for *fixed* edge multiplicities [3] is related to neutral-fibre navigation, whereas the present obstruction concerns descent of an explicit quadratic multiplicity defect. Recent approaches to the Berge–Fulkerson conjecture through partial coverings are also distinct [4].

The main new candidate contribution is the sharp realized unit-defect coordination hierarchy $Q_2 = 6$, $Q_3 = 12$, $Q_4 = 40$, alongside the certified one-coordinate plateau and lifting

results. Whether this qualifies for a research note requires a targeted literature audit and independent external specialist review. Nothing here proves Berge–Fulkerson for nonbipartite cubic graphs. Promising follow-ups include determining the minimum coordinate width for strict descent in nonbipartite graph classes, and using small dual certificates as Lean-checkable obstruction witnesses.

Reproducibility. All graph edges, ordered matching masks, six integer dual potentials, the original and new stand-alone verifiers, the $K_{3,2}$ lift implementation, the covering construction, and regression tests accompany this draft. The current environment contains no Lean 4 installation; formalization has therefore not been completed or claimed.

Research-process disclosure. Computer-assisted search, certificate generation, proof exploration and drafting used ChatGPT (GPT-6); an independently checkable proof record is provided, and the draft remains subject to external review.

References

[1] D. Kőnig, *Gráfok és alkalmazásuk a determinánsok és a halmazok elméletére*, Matematikai és Természettudományi Értesítő **34** (1916), 104–119.

[2] M. Bonamy, N. Bousquet, M. Heinrich, T. Ito, Y. Kobayashi, A. Mary, M. Mühlenthaler and K. Wasa, *The perfect matching reconfiguration problem*, MFCS 2019, LIPIcs **138**, Art. 80. doi:10.4230/LIPIcs.MFCS.2019.80.

[3] K. Mori, R. Motomura, H. Ohsugi and A. Tsuchiya, *Toric ideals of matching polytopes and edge colorings*, Mathematika (2026). doi:10.1112/mtk.70099.

[4] Y. Luo, R.-X. Hao, R. Luo, C.-Q. Zhang and W. Zhou, *Berge–Fulkerson conjecture, perfect matching partial coverings and odd dividers*, Journal of Graph Theory (2026). doi:10.1002/jgt.70097.